

18. $y = f^{-1}(x)$, $x = f(y) = \sqrt{y+3}$ for $y \geq -3$, $y = f^{-1}(x) = x^2 - 3$ for $x \geq 0$.

19. $y = f^{-1}(x)$, $x = f(y) = -\sqrt{3-2y}$ for $y \leq 3/2$, $y = f^{-1}(x) = (3-x^2)/2$ for $x \leq 0$.

20. $y = f^{-1}(x)$, $x = f(y) = y - 5y^2$ for $y \geq 1$, $y = f^{-1}(x) = (1 + \sqrt{1-20x})/10$ for $x \leq -4$.

21. $y = f^{-1}(x)$, $x = f(y) = ay^2 + by + c$, $ay^2 + by + c - x = 0$, use the quadratic formula to get $y = \frac{-b \pm \sqrt{b^2 - 4a(c-x)}}{2a}$;

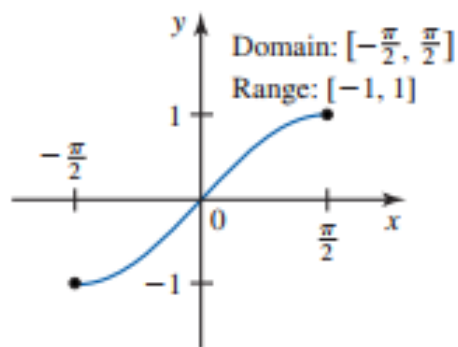
(a) $f^{-1}(x) = \frac{-b + \sqrt{b^2 - 4a(c-x)}}{2a}$

(b) $f^{-1}(x) = \frac{-b - \sqrt{b^2 - 4a(c-x)}}{2a}$

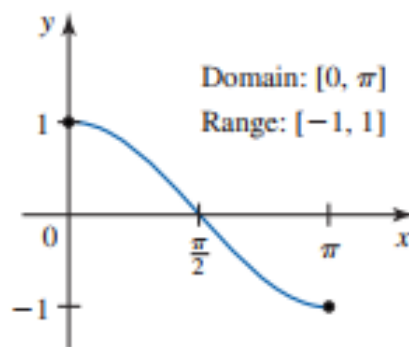
22. (a) $C = \frac{5}{9}(F - 32)$.

(b) How many degrees Celsius given the Fahrenheit temperature.

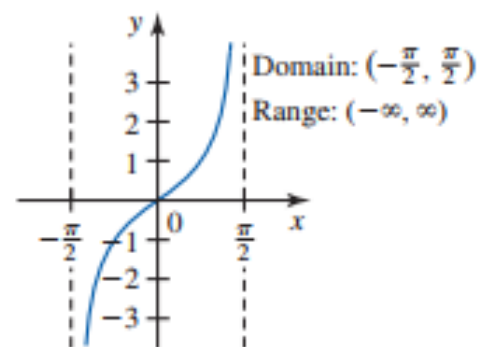
(c) $C = -273.15^\circ \text{C}$ is equivalent to $F = -459.67^\circ \text{F}$, so the domain is $F \geq -459.67$, the range is $C \geq -273.15$.



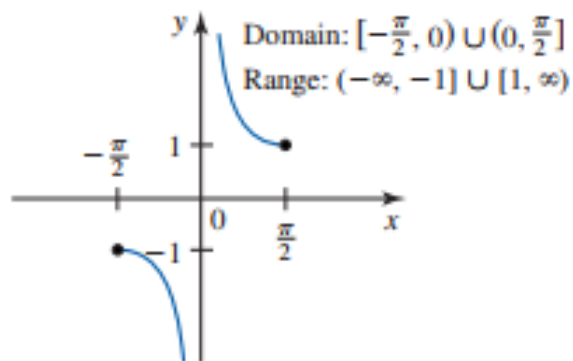
(a) $y = \sin x$



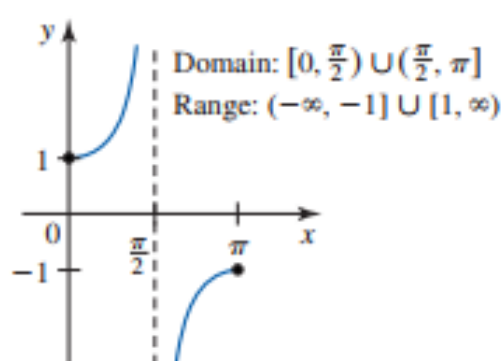
(b) $y = \cos x$



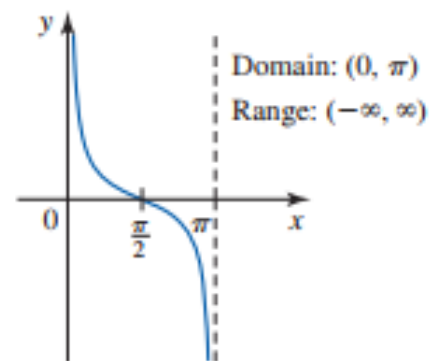
(c) $y = \tan x$



(d) $y = \csc x$

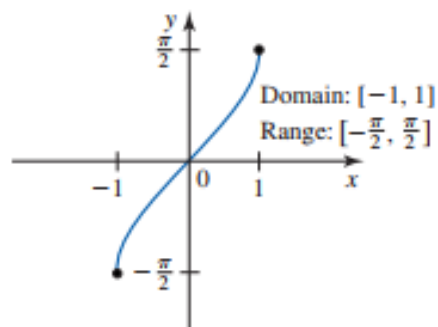


(e) $y = \sec x$

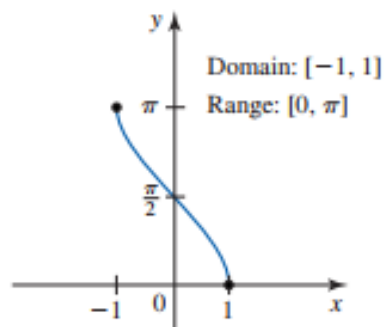


(f) $y = \cot x$

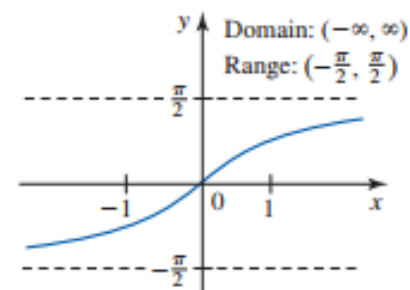
FIGURE 12



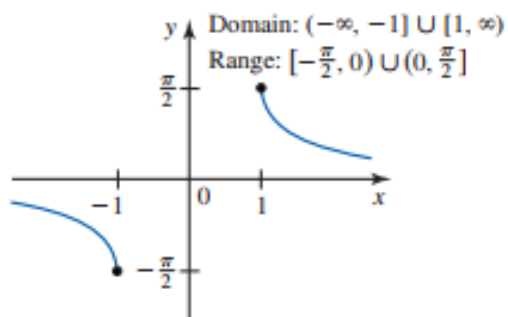
(a) $y = \sin^{-1} x$



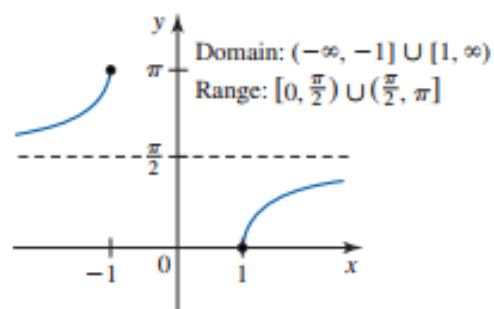
(b) $y = \cos^{-1} x$



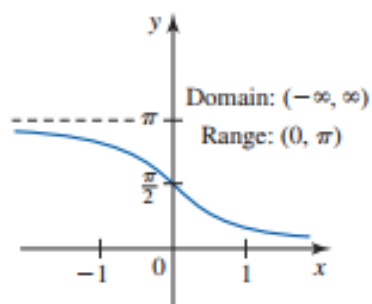
(c) $y = \tan^{-1} x$



(d) $y = \csc^{-1} x$

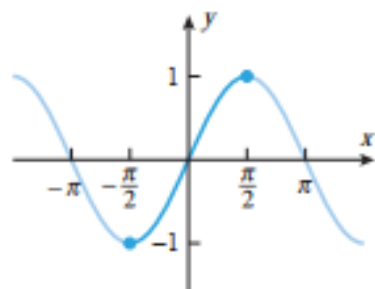


(e) $y = \sec^{-1} x$



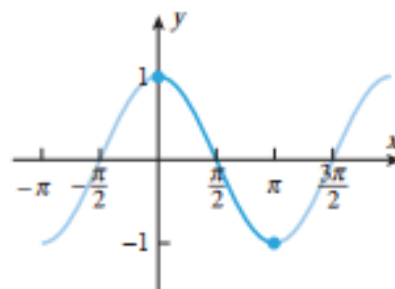
(f) $y = \cot^{-1} x$

FIGURE 13



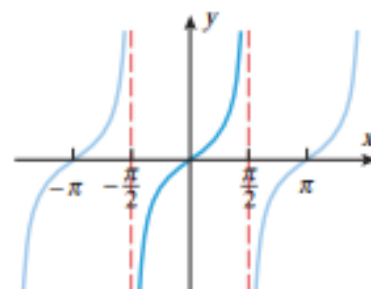
$$y = \sin x$$

$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$



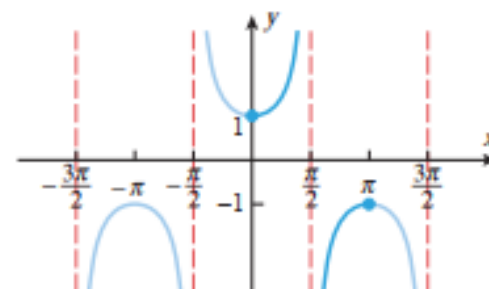
$$y = \cos x$$

$$0 \leq x \leq \pi$$



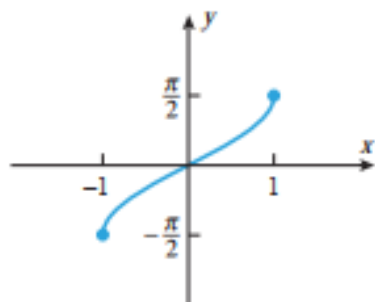
$$y = \tan x$$

$$-\frac{\pi}{2} < x < \frac{\pi}{2}$$

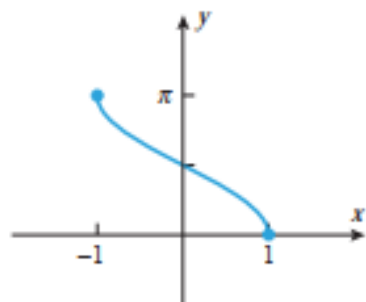


$$y = \sec x$$

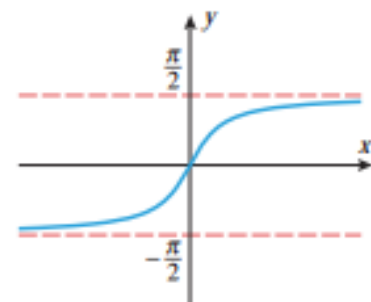
$$0 \leq x \leq \pi, x \neq \frac{\pi}{2}$$



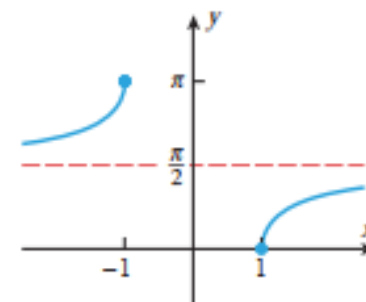
$$y = \sin^{-1} x$$



$$y = \cos^{-1} x$$



$$y = \tan^{-1} x$$

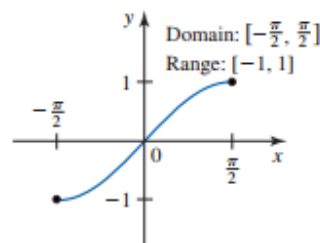


$$y = \sec^{-1} x$$

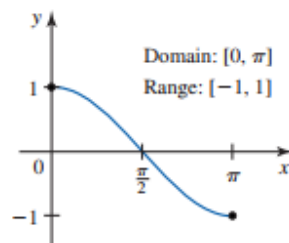
Table 0.4.1

PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

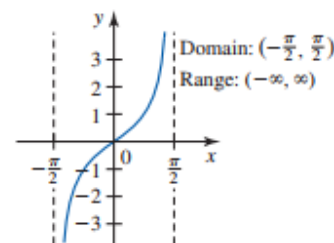
FUNCTION	DOMAIN	RANGE	BASIC RELATIONSHIPS
$\sin^{-1}$	$[-1, 1]$	$[-\pi/2, \pi/2]$	$\sin^{-1}(\sin x) = x$ if $-\pi/2 \leq x \leq \pi/2$ $\sin(\sin^{-1} x) = x$ if $-1 \leq x \leq 1$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$	$\cos^{-1}(\cos x) = x$ if $0 \leq x \leq \pi$ $\cos(\cos^{-1} x) = x$ if $-1 \leq x \leq 1$
$\tan^{-1}$	$(-\infty, +\infty)$	$(-\pi/2, \pi/2)$	$\tan^{-1}(\tan x) = x$ if $-\pi/2 < x < \pi/2$ $\tan(\tan^{-1} x) = x$ if $-\infty < x < +\infty$
$\sec^{-1}$	$(-\infty, -1] \cup [1, +\infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$	$\sec^{-1}(\sec x) = x$ if $0 \leq x \leq \pi, x \neq \pi/2$ $\sec(\sec^{-1} x) = x$ if $ x \geq 1$



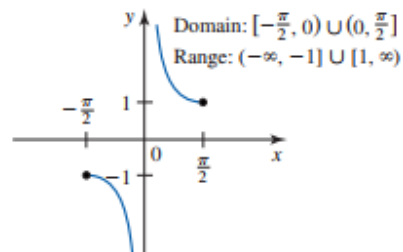
(a) $y = \sin x$



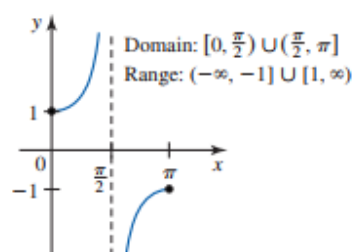
(b) $y = \cos x$



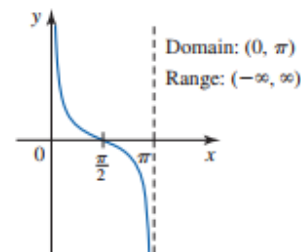
(c) $y = \tan x$



(d) $y = \csc x$

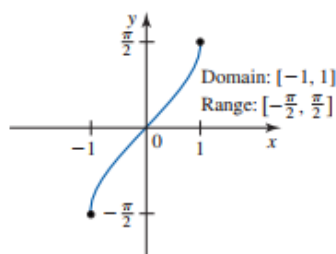


(e) $y = \sec x$

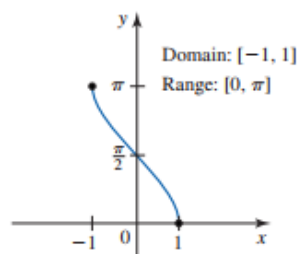


(f) $y = \cot x$

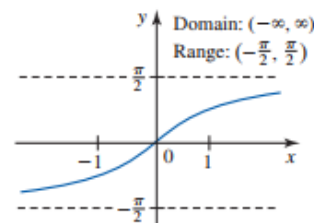
FIGURE 12



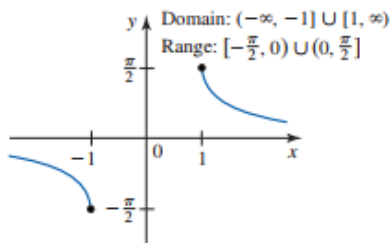
(a) $y = \sin^{-1} x$



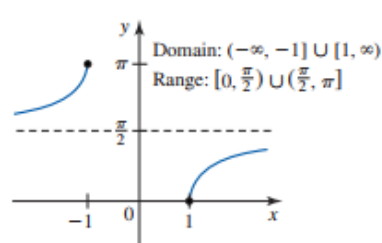
(b) $y = \cos^{-1} x$



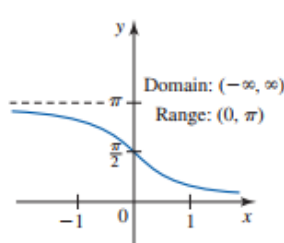
(c) $y = \tan^{-1} x$



(d) $y = \csc^{-1} x$



(e) $y = \sec^{-1} x$



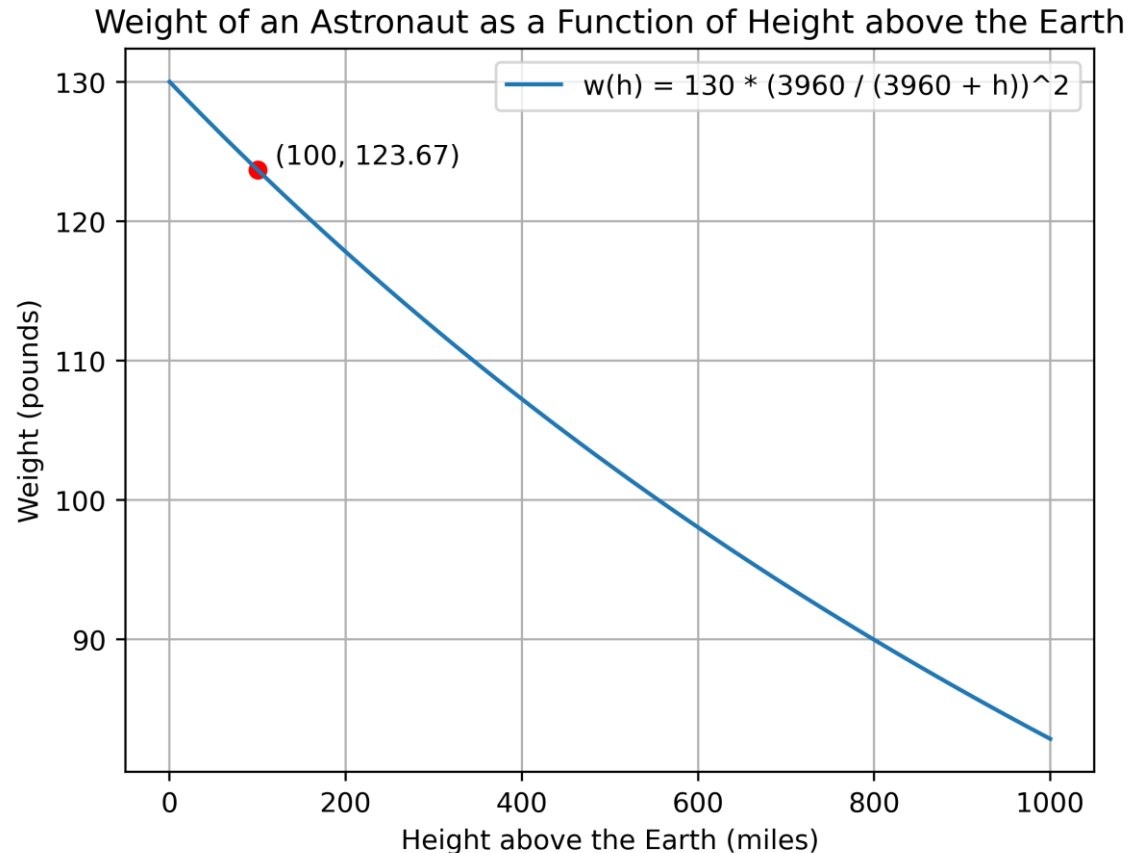
(f) $y = \cot^{-1} x$

FIGURE 13

If an astronaut weighs 130 pounds on the surface of the earth, then her weight when she is h miles above the earth is given by the function:

$$w(h) = 130 \left(\frac{3960}{3960 + h} \right)^2.$$

- i. What is her weight when she is 100 miles above the earth?
- ii. Construct a table of values for the function w that gives her weight at heights from 0 to 500 miles. What do you conclude from the table? Graph the results



Ex-6: If an astronaut weighs 130 pounds on the surface of the earth, then her weight when she is h miles above the earth is given by the function

$$W(h) = 130 \left(\frac{3960}{3960+h} \right)^2$$

(a) What is her weight when she is 100 miles above the earth?

(c) Construct a table of values for the function W that gives her weight at heights from 0 to 500 miles. What do you conclude from the

Ans. $W(100) \approx 123.67$

The higher the astronaut travels,
the less she weighs.

EXAMPLE 3 Using an Exponential Function to Model Population Growth

According to the U.S. Census Bureau, the world population reached 6 billion people during 1999 and was growing exponentially. By the end of 2010, the population had grown to 6.947 billion. The projected world population (in billions of people) t years after 2010 is given by the function

$$f(t) = 6.947e^{0.00745t}.$$

(a) Based on this model, what will the world population be in 2025?

(b) If this trend continues, approximately when will the world population reach 9 billion?

SOLUTION

(a) Since $t = 0$ represents the year 2010, in 2025, t would be $2025 - 2010 = 15$ yr. We must find $f(t)$ when t is 15.

$$f(t) = 6.947e^{0.00745t} \quad \text{Given function}$$

$$f(15) = 6.947e^{0.00745(15)} \quad \text{Let } t = 15.$$

$$f(15) \approx 7.768 \quad \text{Use a calculator.}$$

The population will be 7.768 billion at the end of 2025.

$$(b) \quad f(t) = 6.947e^{0.00745t} \quad \text{Given function}$$

$$9 = 6.947e^{0.00745t} \quad \text{Let } f(t) = 9.$$

$$\frac{9}{6.947} = e^{0.00745t} \quad \text{Divide by 6.947.}$$

$$\ln \frac{9}{6.947} = \ln e^{0.00745t} \quad \text{Take the natural logarithm on each side.}$$

$$\ln \frac{9}{6.947} = 0.00745t \quad \text{In } e^x = x, \text{ for all } x.$$

$$t = \frac{\ln \frac{9}{6.947}}{0.00745} \quad \text{Divide by 0.00745 and rewrite.}$$

$$t \approx 34.8 \quad \text{Use a calculator.}$$

Thus, 34.8 yr after 2010, during the year 2044, world population will reach 9 billion.

EXAMPLE 6 Modeling Newton's Law of Cooling

Newton's law of cooling says that the rate at which a body cools is proportional to the difference in temperature between the body and the environment around it. The temperature $f(t)$ of the body at time t in appropriate units after being introduced into an environment having constant temperature T_0 is

$$f(t) = T_0 + Ce^{-kt}, \text{ where } C \text{ and } k \text{ are constants.}$$

A pot of coffee with a temperature of 100°C is set down in a room with a temperature of 20°C . The coffee cools to 60°C after 1 hr.

(a) Write an equation to model the data.

(b) Find the temperature after half an hour.

(c) How long will it take for the coffee to cool to 50°C ?

SOLUTION

- (a) We must find values for C and k in the given formula. As given, when $t = 0$, $T_0 = 20$, and the temperature of the coffee is $f(0) = 100$.

$$f(t) = T_0 + Ce^{-kt} \quad \text{Given function}$$

$$100 = 20 + Ce^{-0k} \quad \text{Let } t = 0, f(0) = 100, \text{ and } T_0 = 20.$$

$$100 = 20 + C \quad e^0 = 1$$

$$80 = C \quad \text{Subtract 20.}$$

The following function models the data.

$$f(t) = 20 + 80e^{-kt} \quad \text{Let } T_0 = 20 \text{ and } C = 80.$$

The coffee cools to 60°C after 1 hr, so when $t = 1$, $f(1) = 60$.

$$f(t) = 20 + 80e^{-kt} \quad \text{Above function with } T_0 = 20 \text{ and } C = 80$$

$$60 = 20 + 80e^{-1k} \quad \text{Let } t = 1 \text{ and } f(1) = 60.$$

$$40 = 80e^{-k} \quad \text{Subtract 20.}$$

$$\frac{1}{2} = e^{-k} \quad \text{Divide by 80.}$$

$$\ln \frac{1}{2} = \ln e^{-k} \quad \text{Take the natural logarithm on each side.}$$

$$\ln \frac{1}{2} = -k \quad \text{In } e^x = x, \text{ for all } x.$$

$$k \approx 0.693 \quad \text{Multiply by } -1, \text{ rewrite, and use a calculator.}$$

Thus, the model is $f(t) = 20 + 80e^{-0.693t}$.

- (b) To find the temperature after $\frac{1}{2}$ hr, let $t = \frac{1}{2}$ in the model from part (a).

$$f(t) = 20 + 80e^{-0.693t} \quad \text{Model from part (a)}$$

$$f\left(\frac{1}{2}\right) = 20 + 80e^{(-0.693)(1/2)} \quad \text{Let } t = \frac{1}{2}.$$

$$f\left(\frac{1}{2}\right) \approx 76.6^\circ\text{C} \quad \text{Use a calculator.}$$

- (c) To find how long it will take for the coffee to cool to 50°C , let $f(t) = 50$.

$$f(t) = 20 + 80e^{-0.693t} \quad \text{Model from part (a)}$$

$$50 = 20 + 80e^{-0.693t} \quad \text{Let } f(t) = 50.$$

$$30 = 80e^{-0.693t} \quad \text{Subtract 20.}$$

$$\frac{3}{8} = e^{-0.693t} \quad \text{Divide by 80.}$$

$$\ln \frac{3}{8} = \ln e^{-0.693t} \quad \text{Take the natural logarithm on each side.}$$

$$\ln \frac{3}{8} = -0.693t \quad \text{In } e^x = x, \text{ for all } x.$$

$$t = \frac{\ln \frac{3}{8}}{-0.693} \quad \text{Divide by } -0.693 \text{ and rewrite.}$$

$$t \approx 1.415 \text{ hr, or about 1 hr, 25 min} \quad \text{Now Try Exercise 27.}$$

56. Over-100 Population On the basis of data obtained from the Census Bureau, the number of Americans over age 100 years is expected to be

$$P(t) = 0.07e^{0.54t} \quad 0 \leq t \leq 4$$

where $P(t)$ is measured in millions and t is measured in decades, with $t = 0$ corresponding to the beginning of 2000. What was the population of Americans over age 100 years at the beginning of 2000? What will it be at the beginning of 2030?

This corresponds to $t = 0$:

$$P(0) = 0.07e^{0.54 \cdot 0} = 0.07e^0 = 0.07 \text{ million} = \boxed{70,000}$$

Answer: At the beginning of 2000, the population over age 100 was **70,000**.

The year 2030 is **3 decades** after 2000, so $t = 3$:

$$P(3) = 0.07e^{0.54 \cdot 3} = 0.07e^{1.62}$$

Calculate:

$$e^{1.62} \approx 5.053 \Rightarrow P(3) \approx 0.07 \times 5.053 \approx 0.3537 \text{ million}$$

$$P(3) \approx \boxed{353,700}$$

Answer: At the beginning of 2030, the population over age 100 is expected to be approximately **353,700**.

A cheesecake is taken out of the oven with an ideal internal temperature of $165^{\circ}F$, and is placed into a $35^{\circ}F$ refrigerator. After 10 minutes, the cheesecake has cooled to $150^{\circ}F$. If we must wait until the cheesecake has cooled to $70^{\circ}F$ before we eat it, how long will we have to wait?

Solution

Because the surrounding air temperature in the refrigerator is 35 degrees, the cheesecake's temperature will decay exponentially toward 35, following the equation

$$T(t) = D_0 e^{-kt} + 35$$

We know the initial temperature was 165, so $T(0) = 165$.

$$\begin{aligned} 165 &= A e^{-k \cdot 0} + 35 && \text{Substitute } (0, 165) \\ D_0 &= 130 && \text{Solve for } D_0 \end{aligned}$$

We were given another data point, $T(10) = 150$, which we can use to solve for k .

$$\begin{aligned} 150 &= 130 e^{-k \cdot 10} + 35 && \text{Substitute } (10, 150) \\ 115 &= 130 e^{-k \cdot 10} && \text{Subtract 35} \\ \frac{115}{130} &= e^{-10k} && \text{Divide by 130} \\ \ln\left(\frac{115}{130}\right) &= -10k && \text{Take the natural log of both sides} \\ k &= \frac{\ln\left(\frac{115}{130}\right)}{-10} && \text{Divide by the of } k \end{aligned}$$

This gives us the equation for the cooling of the cheesecake: $T(t) = 130 e^{\left(-\frac{1}{10} \ln \frac{23}{26} t\right)} + 35$.

Now we can solve for the time it will take for the temperature to cool to 70 degrees.

$$70 = 130e^{\left(-\frac{1}{10}\ln\frac{23}{26}\right)t} + 35 \quad \text{Substitute in 70 for } T(t)$$

$$35 = 130e^{\left(-\frac{1}{10}\ln\frac{23}{26}\right)t} \quad \text{Subtract 35}$$

$$\frac{35}{130} = e^{\left(-\frac{1}{10}\ln\frac{23}{26}\right)t} \quad \text{Divide by 130}$$

$$\ln\left(\frac{35}{130}\right) = \left(-\frac{1}{10}\ln\frac{23}{26}\right)t \quad \text{Take the natural log of both sides}$$

$$t = \frac{\ln\left(\frac{35}{130}\right)}{\left(-\frac{1}{10}\ln\frac{23}{26}\right)}$$

$$t \approx 107 \quad \text{Divide by the of } t$$

It will take about 107 minutes, or one hour and 47 minutes, for the cheesecake to cool to $70^\circ F$.

